

CS-201 DATA STRUCTURES AND ALGORITHMS

Course Syllabus - Semester Outline

TOPIC 1: FOUNDATIONS OF DATA STRUCTURES

1.1 Complexity Analysis

CS201.1.1 Analyse the time complexity of an algorithm using Big-O notation.

CS201.1.2 Compare best-case, average-case and worst-case complexity.

1.2 Arrays and Linked Lists

CS201.2.1 Implement singly and doubly linked list operations.

CS201.2.2 Contrast array and linked-list performance for insertion and deletion.

TOPIC 2: LINEAR STRUCTURES

2.1 Stacks and Queues

CS201.3.1 Implement a stack using arrays and linked lists.

CS201.3.2 Apply stacks to expression evaluation and recursion unwinding.

CS201.3.3 Implement circular and priority queues.

Prerequisite: CS201.2.1

TOPIC 3: TREES

3.1 Binary Trees and Binary Search Trees

CS201.4.1 Perform inorder, preorder and postorder traversal of a binary tree.

CS201.4.2 Insert, search and delete nodes in a binary search tree.

Prerequisite: CS201.2.1

3.2 Balanced Trees

CS201.5.1 Construct an AVL tree and apply rotations to restore balance.

CS201.5.2 Construct a B-tree of a given order and perform insertion.

CS201.5.3 Delete a key from a B-tree and explain the rebalancing process.

Prerequisite: CS201.4.2

TOPIC 4: HASHING

4.1 Hash Tables

CS201.6.1 Design a hash function and evaluate its distribution quality.

CS201.6.2 Resolve collisions using linear probing and quadratic probing.

CS201.6.3 Resolve collisions using separate chaining.

CS201.6.4 Implement hard deletion of keys from an open-addressed hash table.

Prerequisite: CS201.1.1

TOPIC 5: GRAPHS

5.1 Graph Representation and Traversal

CS201.7.1 Represent a graph using adjacency matrix and adjacency list.

CS201.7.2 Traverse a graph using breadth-first and depth-first search.

Prerequisite: CS201.3.3

5.2 Shortest Path and Spanning Trees

CS201.8.1 Compute single-source shortest paths using Dijkstra's algorithm.

CS201.8.2 Construct a minimum spanning tree using Kruskal's algorithm.

Prerequisite: CS201.7.2

TOPIC 6: SORTING ALGORITHMS

6.1 Comparison Sorts

CS201.9.1 Implement merge sort and analyse its complexity.

CS201.9.2 Implement quick sort and discuss pivot selection strategies.

CS201.9.3 Implement heap sort using a binary heap.

Prerequisite: CS201.1.1

ASSESSMENT

Mid-term examinations: 30 percent

Final examination: 50 percent

Assignments and laboratory work: 20 percent